

Please write clearly in block capitals.

Centre number

Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE**Paper 1 (Generated practice paper - Set 23)**

Assessed: The physical environment, Energy resources, Pollution, Research methods

Time allowed: 3 hours**Materials**

For this paper you may use:

- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
 - use good English
 - organise information clearly
 - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
11	
TOTAL	

Answer all questions in the spaces provided.

0 1

A low-lying coastal town in eastern England is at risk of flooding from rising sea level and more intense rainfall.

0 1

1

Describe **three** ways the town could adapt to the increased flood risk.

[3 marks]

1

2

3

0 1

2

Explain the circumstances in which managed retreat may be chosen instead of coastal defences.

[2 marks]

0 1

3

Describe **two** impacts of higher temperatures on transport infrastructure.

[2 marks]

1

2

0 1

4

Explain how carbon capture and storage (CCS) could reduce CO₂ emissions from a fossil fuel power station, and give **one** limitation.

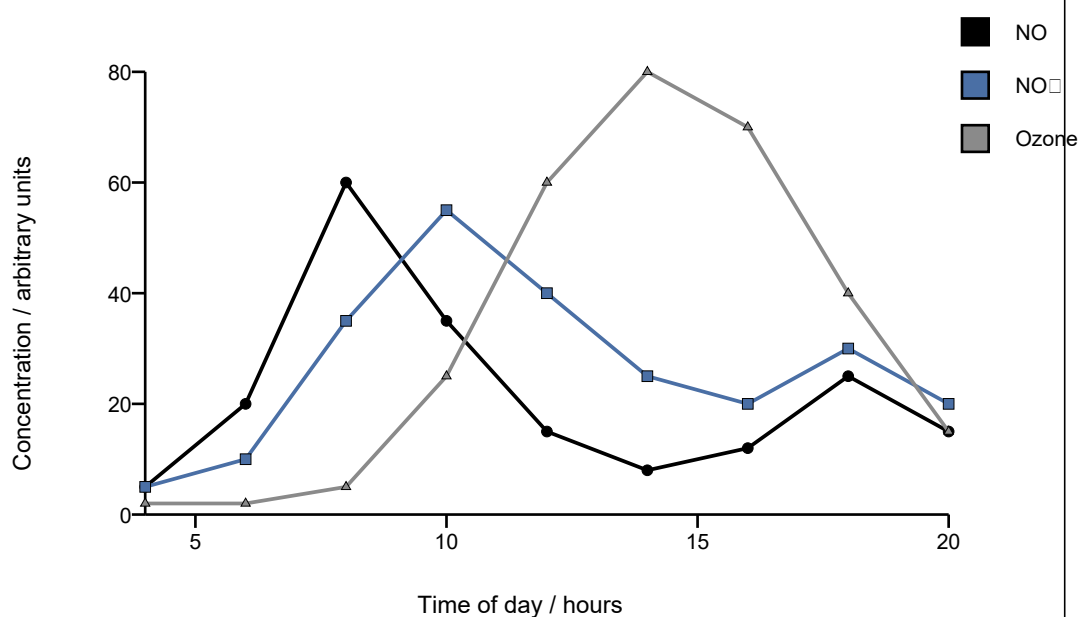
[3 marks]

10

0 2

Figure 1 shows the concentrations of three pollutants in the air of a large city in a valley during one sunny day.

Figure 1



0 2

. 1

Use Figure 1 to describe how the ozone concentration changes during the day and explain the pattern.

[4 marks]

0 2 . 2

Explain why the early-morning peak in NO is followed by a peak in NO₂.**[2 marks]**

0 2 . 3

Name **one** other toxic secondary pollutant produced in photochemical smogs and give one of its effects on humans.**[2 marks]**

0 2 . 4

Explain why cities such as Los Angeles and Mexico City are particularly prone to photochemical smogs.

[2 marks]

10

0 3 . 1

Explain how occupancy sensors and programmable thermostats reduce energy waste.

[2 marks]

0 3 . 2

Explain why LED lights use much less electricity than the filament and CFL lamps they replaced, and give **two** other examples of low-energy appliance design.**[3 marks]**

0 3 . 3

Describe **three** changes in human behaviour that reduce domestic energy use.

[3 marks]

1

2

3

0 3 . 4

Explain why heating water on demand saves energy compared with storing hot water.

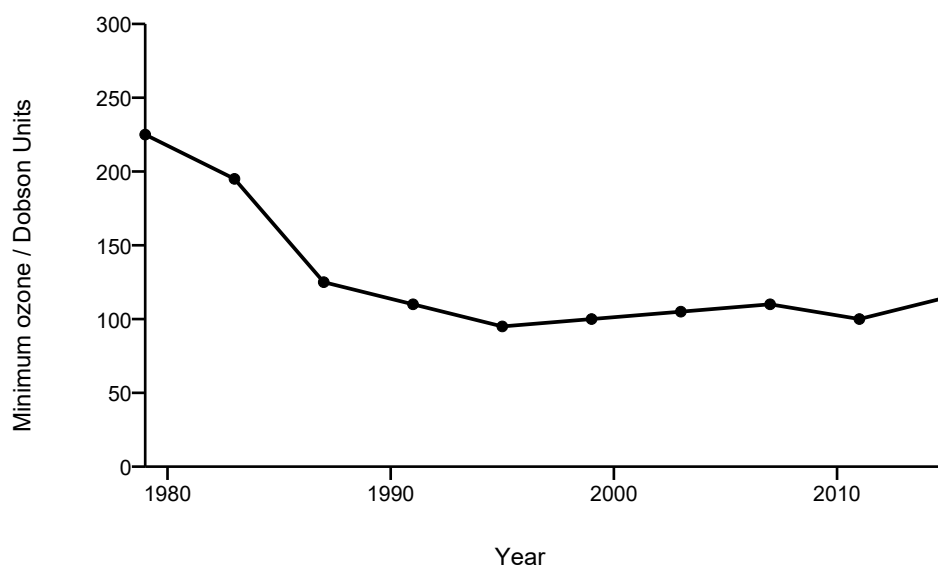
[2 marks]

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10

0 4

Figure 1 shows the minimum ozone concentration recorded over Antarctica each spring between 1979 and 2015.

Figure 1**0 4****1**

Describe the trend shown in Figure 1.

[2 marks]**0 4****2**

Calculate the percentage decrease in minimum ozone between 1979 and 1995.

[2 marks]**Answer**

%**0 4****3**

Explain why ozone depletion is most severe over Antarctica in spring.

[4 marks]

0 4 . 4

Explain why the data in Figure 1 alone could not confirm that CFCs were the cause of the depletion, and describe how the cause was confirmed.

[2 marks]

10

0 5 . 1

Which of the following is a feature of a river that affects its usefulness as a water source? Tick **one** box.

[1 mark]

A Its total annual discharge

☐

B The albedo of its surface

☐

C Its salinity

☐

D The age of its bedrock

☐

0 5 . 2

Explain why rainwater collection is increasingly important in cities such as those in India.

[2 marks]

0 5 . 3

Explain why an estuary barrage has fewer land-use conflicts than a reservoir on land but still causes environmental problems.

[2 marks]

5

0 6 . 1

Explain why nitrates in drinking water can cause **blue baby syndrome** (methaemoglobinaemia).

[3 marks]

0 6 . 2

Explain why the link between nitrates in drinking water and stomach cancer is uncertain.

[2 marks]

0 6 . 3

Describe **three** ways farmers in a Nitrate Vulnerable Zone can reduce nitrate leaching into aquifers.

[3 marks]

1

2

3

0 6 . 4 Explain how phosphates are removed from sewage effluent and from lakes polluted in the past. **[2 marks]**

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10

0 7 . 1 Complete Table 1 to show a suitable sampling technique for each group of organisms and one limitation of the technique. **[6 marks]**

Table 1

Organisms	Technique	One limitation	
Invertebrates active on the ground surface at night			
Invertebrates living in leaf litter			
Invertebrates on tree branches			

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07 . 2

Explain why a light trap cannot be used to prove that a moth species is absent from a site.

[2 marks]

07 . 3

Explain how a motor-driven suction sampler can give a population density, whereas a pooter cannot.

[2 marks]

10

08 . 1

Which of the following properties would make a pesticide **least** likely to biomagnify? Tick one box.

[1 mark]

- A High liposolubility and high persistence
- B Low liposolubility and low persistence
- C High liposolubility and low persistence
- D Low liposolubility and high persistence

☐
☐
☐
☐

08 . 2

Explain the difference between **bioaccumulation** and **biomagnification**.

[2 marks]

08 . 3

Explain why pesticides are described as an unusual type of pollutant.

[2 marks]

5

09

A student determined the water and organic matter content of a soil sample. Table 1 shows the results.

Table 1

Measurement	Mass / g
Empty crucible	25.0
Crucible + fresh soil	65.0
Crucible + soil after drying at 105 °C to constant mass	55.0
Crucible + soil after heating in a furnace at 550 °C	52.6

09 . 1

Calculate the percentage water content of the fresh soil.

[2 marks]

Answer

%

09 . 2

Calculate the percentage organic matter content of the dry soil.

[2 marks]

Answer

%

09 . 3

Explain why the soil is dried to constant mass before the organic matter is burnt off.

[2 marks]

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09 . 4

Explain how the organic matter content affects the fertility and erosion risk of this soil.

[3 marks]

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09 . 5

Suggest **one** reason why two students measuring the same field might get different organic matter results.

[1 mark]

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10

1 0

Table 1 shows the three forms of oxygen found in the stratosphere.

Table 1

Form of oxygen	Number of atoms per molecule
Monatomic oxygen (O)	1
Diatomic oxygen (O ₂)	2
Triatomic oxygen / ozone (O ₃)	3

1 0**1**

Explain how the three forms of oxygen protect life at the surface from ultraviolet radiation.

[3 marks]

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1 0**2**Give **two** other ways in which the atmosphere supports life.**[2 marks]**

1

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2

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1 0**3**

Explain why the atmosphere is thermally stratified into layers.

[1 mark]

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1 0**4**

The troposphere and stratosphere are the layers most affected by human activities. Discuss how human activities alter the energy processes in these two layers and evaluate the difficulties of predicting the consequences.

[9 marks]

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[illegible]

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15

Write an essay on one of the following topics.

$$\boxed{1}\boxed{1}.\boxed{1}$$

Discuss the strategies that can be used to provide sustainable supplies of water for a growing population.

[25 marks]

OR

1 1 . 2

Discuss how the design of a sampling programme and the choice of statistical analysis affect the validity of conclusions drawn from environmental investigations.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

Do not write outside the box

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Do not write outside the box

END OF QUESTIONS